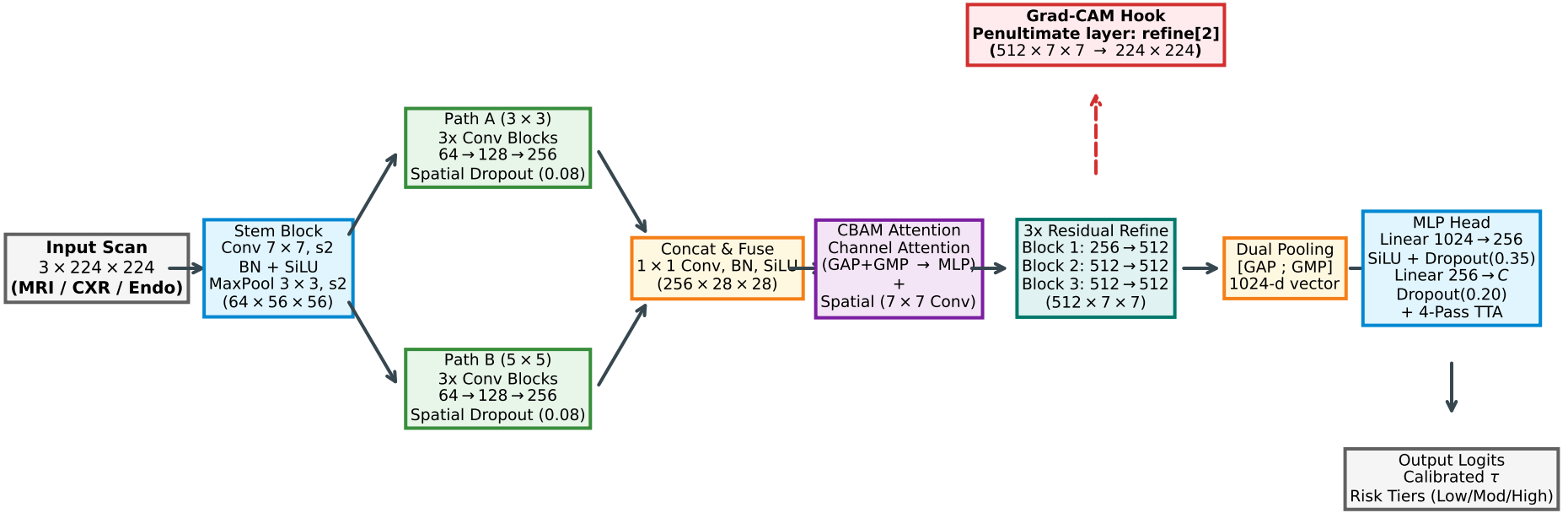
Grad-CAM Hook
Penultimate layer: refine[2]
(512 × 7 × 7 → 224 × 224)
Input Scan
3 × 224 × 224
(MRI / CXR / Endo)
Stem Block
Conv 7 × 7, s2
BN + SiLU
MaxPool 3 × 3, s2
(64 × 56 × 56)
Path A (3 × 3)
3x Conv Blocks
64 → 128 → 256
Spatial Dropout (0.08)
Path B (5 × 5)
3x Conv Blocks
64 → 128 → 256
Spatial Dropout (0.08)
Concat & Fuse
1 × 1 Conv, BN, SiLU
(256 × 28 × 28)
CBAM Attention
Channel Attention
(GAP+GMP → MLP)
+
Spatial (7 × 7 Conv)
3x Residual Refine
Block 1: 256 → 512
Block 2: 512 → 512
Block 3: 512 → 512
(512 × 7 × 7)
Dual Pooling
[GAP ; GMP]
1024-d vector
MLP Head
Linear 1024 → 256
SiLU + Dropout(0.35)
Linear 256 → C
Dropout(0.20)
+ 4-Pass TTA
Output Logits
Calibrated τ
Risk Tiers (Low/Mod/High)